Mind the Gap:
Topology-Aware in 3D Completion

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Code and Benchmark available:
https://github.com/danielye0010/Mind-the-Gap-Topology-3D-Completion.git



Outline

- Motivation
- Problem Statement
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- Proposed Method
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- Future Directions (short & long)

Motivation

Current 3D shape completion models excel at reconstructing visually accurate shapes but often fail to preserve the underlying topological structure. These topological errors, while visually subtle, can lead to critical failures in real-world applications like robotic manipulation and simulation. Moreover, widely-used benchmarks focus solely on geometric accuracy and do not penalize incorrect topology. To truly advance the reliability of 3D completion systems, we need both **topology-aware models** and evaluation **benchmarks** that emphasize structural integrity, not just surface reconstruction.

Problem Statement

To address this, our research is driven by the question:

How can deep learning models for point cloud completion become topology-aware and what are the best ways to incorporate topological signals into such models?

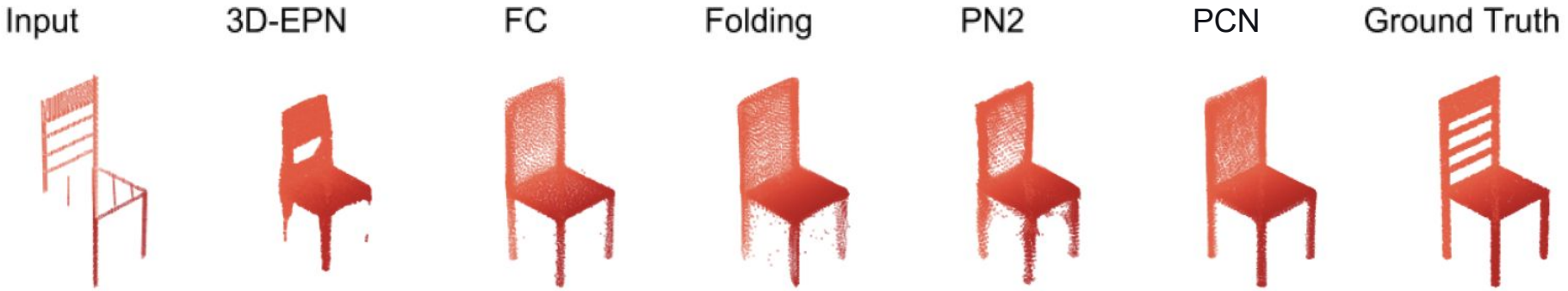
We propose a systematic investigation of this question using two complementary strategies:

1. **Topology-Encoded Data Augmentation / Topology-aware Benchmark**

We inject explicit 2D disk-shaped holes, a topological feature, into both partial and complete point clouds in the training and testing data. Furthermore, we explore augmenting each point cloud with the most persistent feature as additional input channels to guide learning.

2. **Topology-Aware Model**

In addition to Chamfer Distance, we incorporate persistent homology–based loss, which is H1 Bottleneck Distance between persistence diagrams into the training objective. This encourages the model to minimize not only geometric error but also topological discrepancy between predicted and ground truth shapes.



Borrowed from PCN [1]

Goal

The goal of this paper is to advance 3D shape completion by introducing topological awareness into both model design and evaluation. Specifically, we aim to:

1. Develop a **topology-aware benchmark**—ModelNet-Topology—that introduces controlled 1D topological features (holes) into point clouds for evaluating structural fidelity under partial and corrupted inputs.

2. Design and implement **Topology-augmented PCN** (Topo-PCN), a shape completion model that incorporates persistent homology-based features and a topology-guided loss to improve topological correctness during reconstruction.

3. Provide a comprehensive evaluation of Topo-PCN, PCN, and PointAttN to assess their ability to recover both geometric and topological properties of 3D shapes.

Dataset

ModelNet-C (https://pointcloud-c.github.io/home.html)

Total unique labels: 40

Total number of samples: 2468

Each dataset clean: (1024, 3)

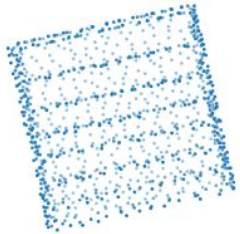
Dropout : (925 ,3)

Dataset

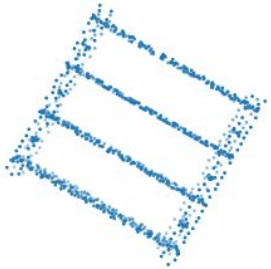
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sample_1182.txt



sample_16.txt



sample_1189.txt



sample_2316.txt



sample_2260.txt



sample_519.txt



sample_1662.txt



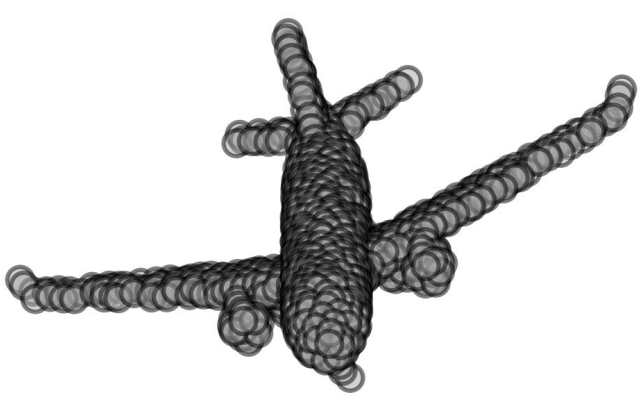
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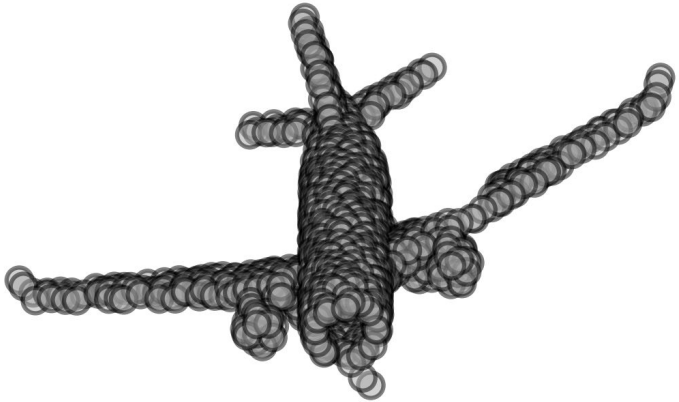
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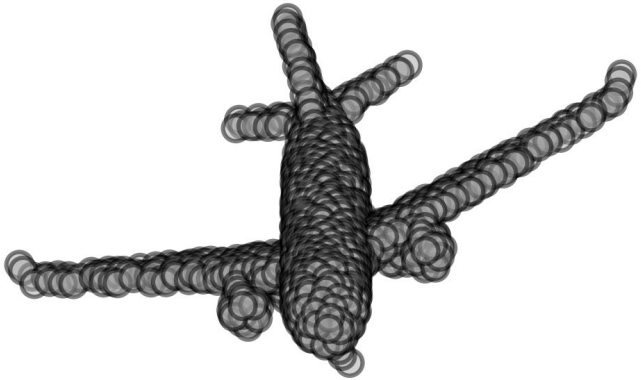


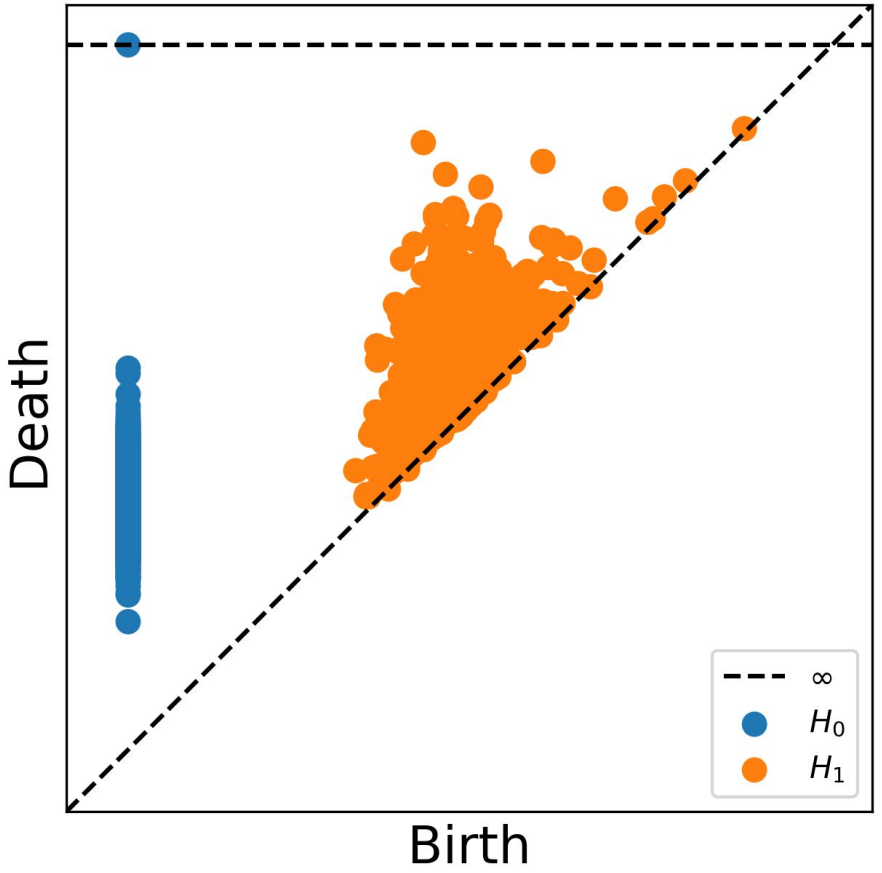
Dataset (AirPlane Complete)

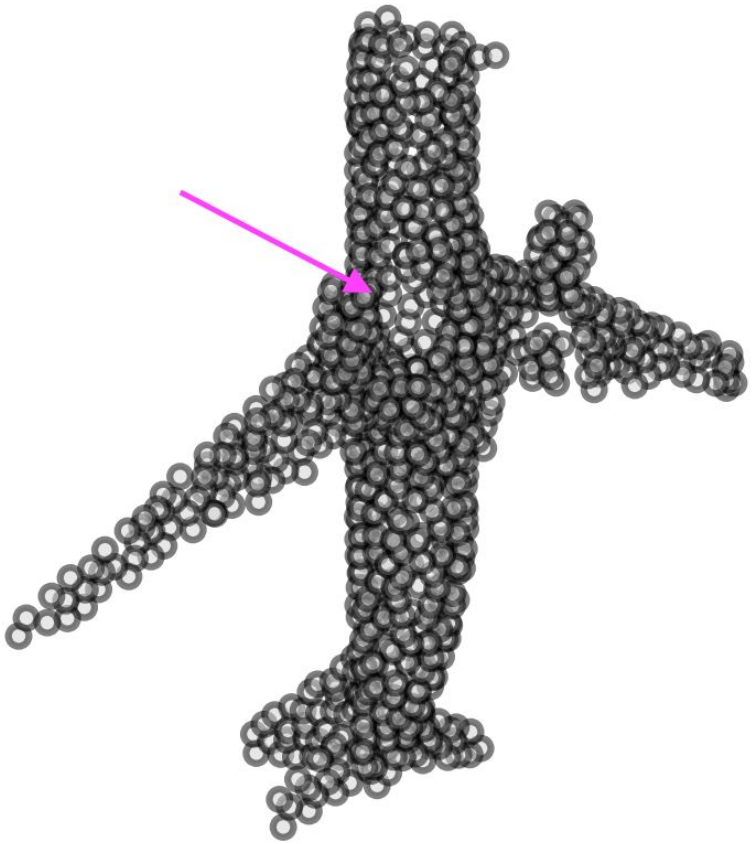


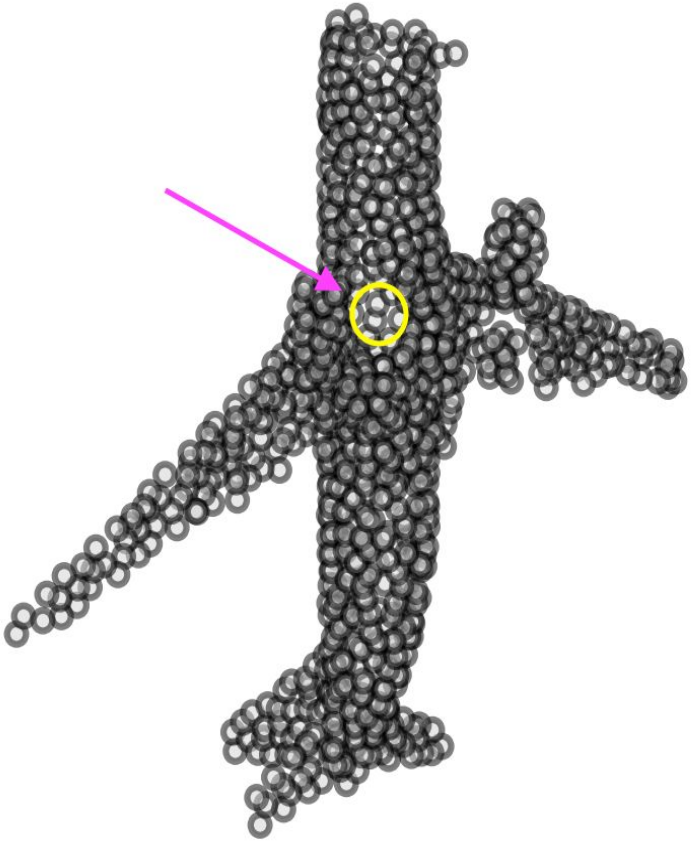
Dataset (AirPlane Partial)

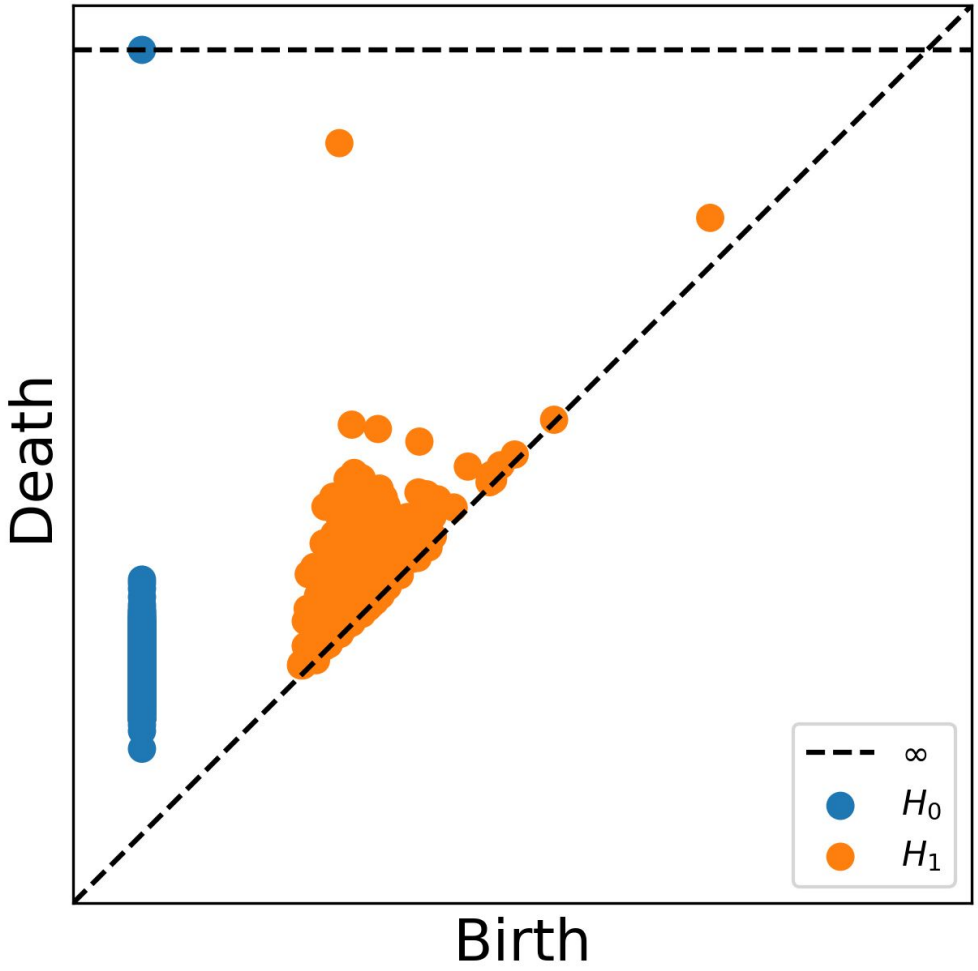





Death
Birth
∞
H_0
H_1






Death
Birth
∞
H_0
H_1

Topology Vector (Birth, Death, Persistence)

Method - Baseline Models

The benchmark dataset provides a new and feasible testbed for 3d completion models to test the topology ability

Baseline: *Point Completion Network (Yuan et al, 3DV 2018):*

Standard 3-layer-MLP encoder-decoder - milestone model in 3d task, simple model

Current SOTA: *PointAttention (Wang et al. AAAI 2024):*

Multi-Head Attention encoder + mlp decoder

Our Method: Topo enhanced version of PCN

(enhanced version of topo + attention on the way)

Method - Training Details

Input: partial plane with a hole, desampled to 1024 points saving time (16000+)

Output: complete plane with the same hole, 4096 points

Same strategy as the original paper:

Loss: Chamfer Distance - A measure of the average closest point distance between two sets of points, calculated by finding the nearest point in one set for each point in the other set and averaging these distances.

8/2 train-test

LR = 1e-4

20 epochs (original is 500/400)

Relu + Adam

TopoPCN - Topology Aware model

Input: Partial Point Cloud (1024 points) **+ Topology Vector (Birth, Death, Persistence)**

= (x,y,z,Birth,Death,Persistence), 6*1024d vector

Output: Complete Point Cloud (4096 points)

Two-Block Encoder:

Block 1: Local Feature Extraction (**6** → 128 → 256), then Max Pooling to obtain a global 256D vector

Block 2: Global Feature Extraction (256 → 512 → 1024), further enhancing global feature

Decoder:

MLP: (1024 → 4096 × 3)

Generates: 4096 3D points

TopoPCN - Topological Loss

Loss: $L = \text{Chamfer Distance} + \lambda \times \text{H1 Bottleneck Distance}$

- Chamfer Distance (CD): Evaluates geometric accuracy between predicted and target point clouds using L2 distance.
- H1 Bottleneck Distance: Quantifies topological similarity between predicted and target point clouds using Persistent Homology, focusing on loops and holes.

Method - Metric

- Chamfer Distance (CD):

Overall geometric accuracy

- Hole Chamfer Distance (Hole CD):

Measures the topological reconstruction accuracy within the hole regions to see how well it recovers the hole

Results

Model	Validation CD	Hole CD
Baseline-PCN	0.001237	0.3699
Topo-PCN	0.001256	0.2962
PointAttention	0.004500	0.1648

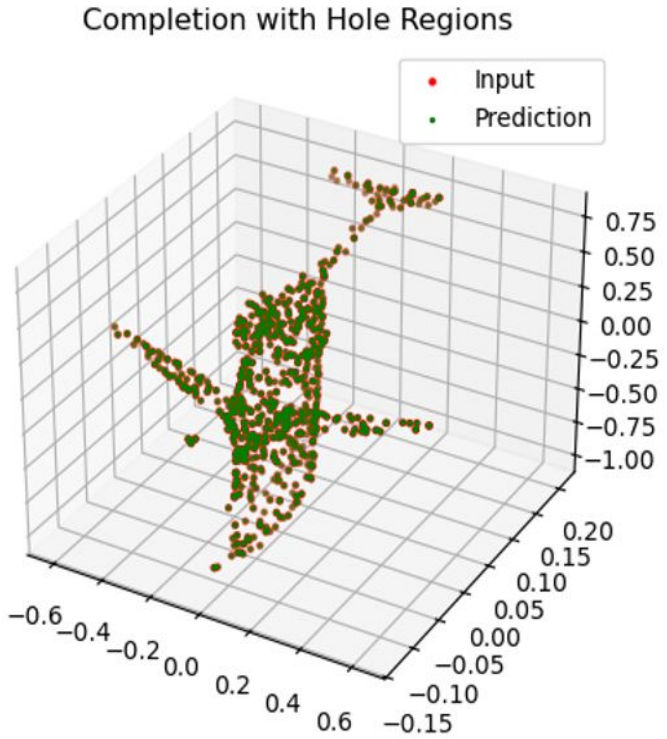
Interpretation

Topo-PCN: Improved topological accuracy (Hole CD: 0.296 vs. 0.369) with minimal impact on geometric accuracy (~0.0012).

PointAttention: Best topological accuracy (Hole CD: 0.164) but lower geometric accuracy (CD: 0.0045) - attention need more data

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Topo-PCN	0.001256	0.2962
PointAttention*	0.004500	0.1648

Completion visualization of TopoPCN



Conclusions

- **Our benchmark** (have the potential to be) **the first one that can effectively evaluate models' topology completion ability. The current ones only for geometry.**

- **Topo enhance works** - Explicitly add topology features and losses directly can enhance topological completion without sacrificing geometric quality, but computing those feature is expensive.
- **Attention is all you need?** - Attention is good at automatically notice the topology, but calculation in point cloud is expensive, and need more training data

- Both ways can enhance topology, but both are slow and expensive
- Looks like **attention** is better in our result
- Future research direction might be fast/efficient attention or topology calculation

Continuing ...

- Update the benchmark with more topology challenge categories and more evaluation metric
- Test more baseline models on our benchmark to see topology ability
- Add more topology features like Cycle Points for explicit **location** guidance in completion.
- Integrate topology with attention-based model

References

[1] *PointAttN: You Only Need Attention for Point Cloud Completion, Wang et al. AAAI 2024*

[2] *PCN: Point Completion Network, Yuan et al, 3DV 2018*

